Evaluating Diffusion Models for Cosmological Simulation Data: Memorization, Generalization, and Scientific Fidelity

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The question

Generative emulators promise **cheap surrogates** for expensive cosmological simulations — but an emulator that **memorizes** its training set produces confident nonsense when used for inference. We train diffusion models on CAMELS HI maps and ask:

1. When does the model stop **copying** training fields and start **sampling** the underlying distribution?
2. Does conditional generation actually **respond to the requested cosmology** θ?
3. Are generated fields **statistically faithful** (one-point PDF, power spectrum)?

Data & models

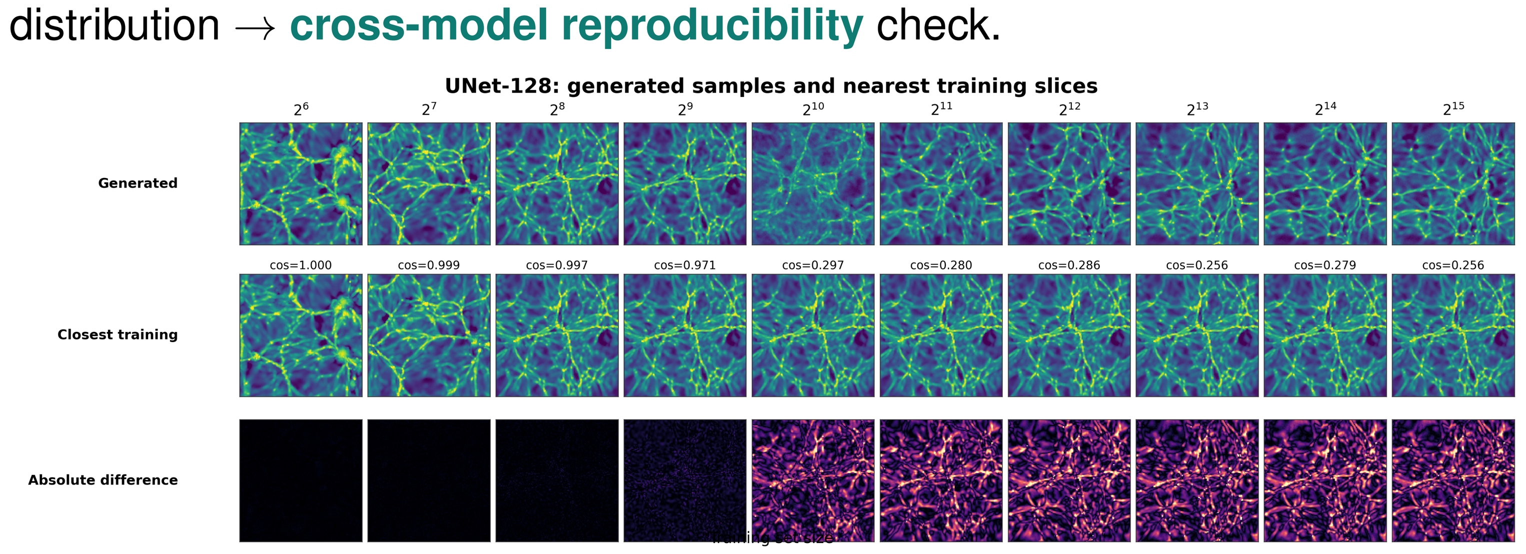
- **CAMELS** (IllustrisTNG) HI fields $\rightarrow$ 128×128 slices; 6 parameters Ω_m, σ_8, A_{SN1}, A_{AGN1}, A_{SN2}, A_{AGN2}.
- **Diffusion**: UNet denoisers (widths 64 / 128 / 256); conditional variant injects $\boldsymbol{\theta}$ at every level via cross-attention.
- **Sampling**: 50-step DPM-Solver, $8\times$ faster than the 500-step DDPM baseline at matched quality.
- **Training-set sweep**: $N = 2^6 \rightarrow 2^{15}$ fields, matched optimizer-update budget; runs on U-M Great Lakes HPC.

Detecting memorization

For each generated sample x_j, find its nearest training field in pixel, PCA, and SSCD embedding space:

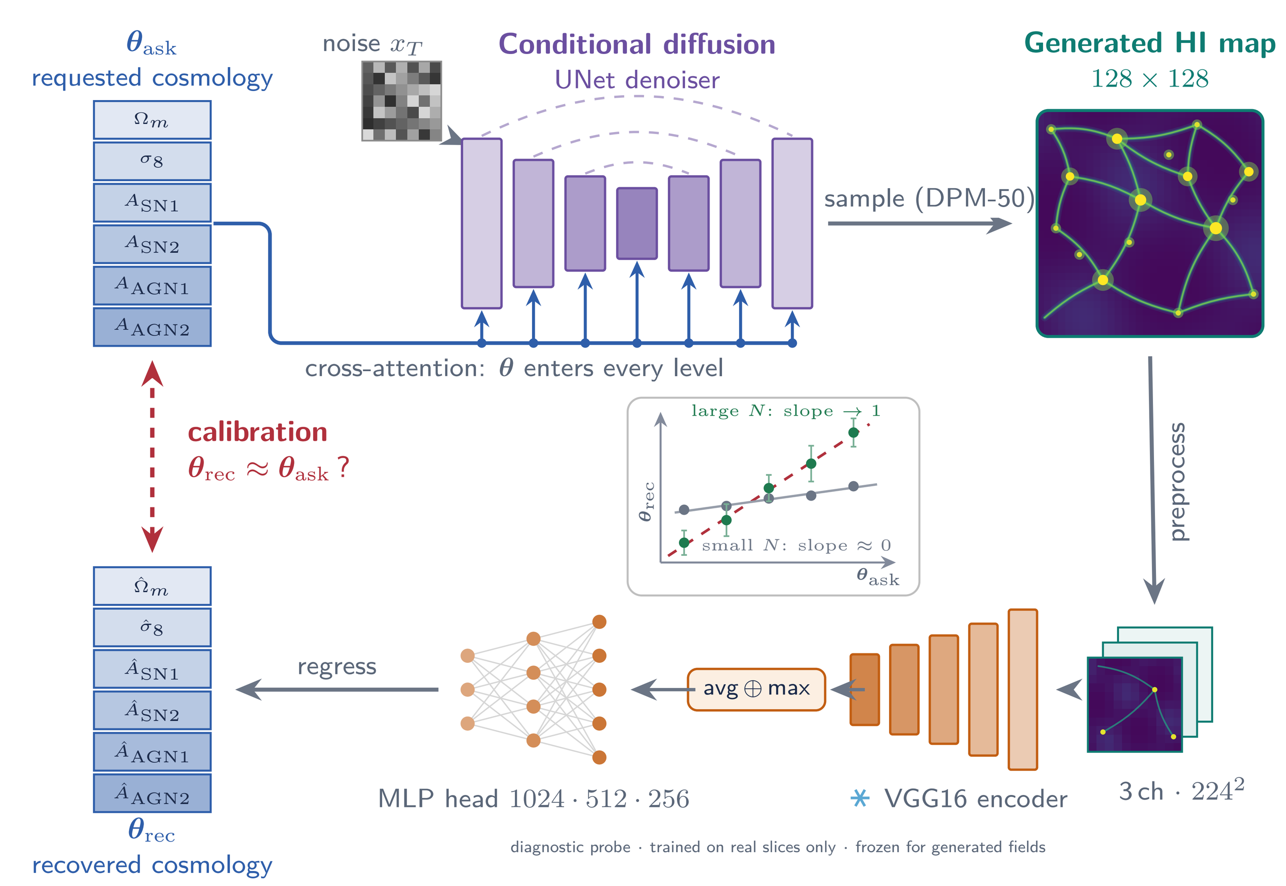
$$s_j = \max_i \mathrm{sim}(x_j, y_i), \qquad \mathrm{GL}(\tau) = 1 - \Pr[s_j > \tau]$$

with τ chosen **adaptively** from leave-one-out train–train similarities. Independent architectures trained on the same data must agree if they learned the same distribution $\rightarrow$ **cross-model reproducibility** check.



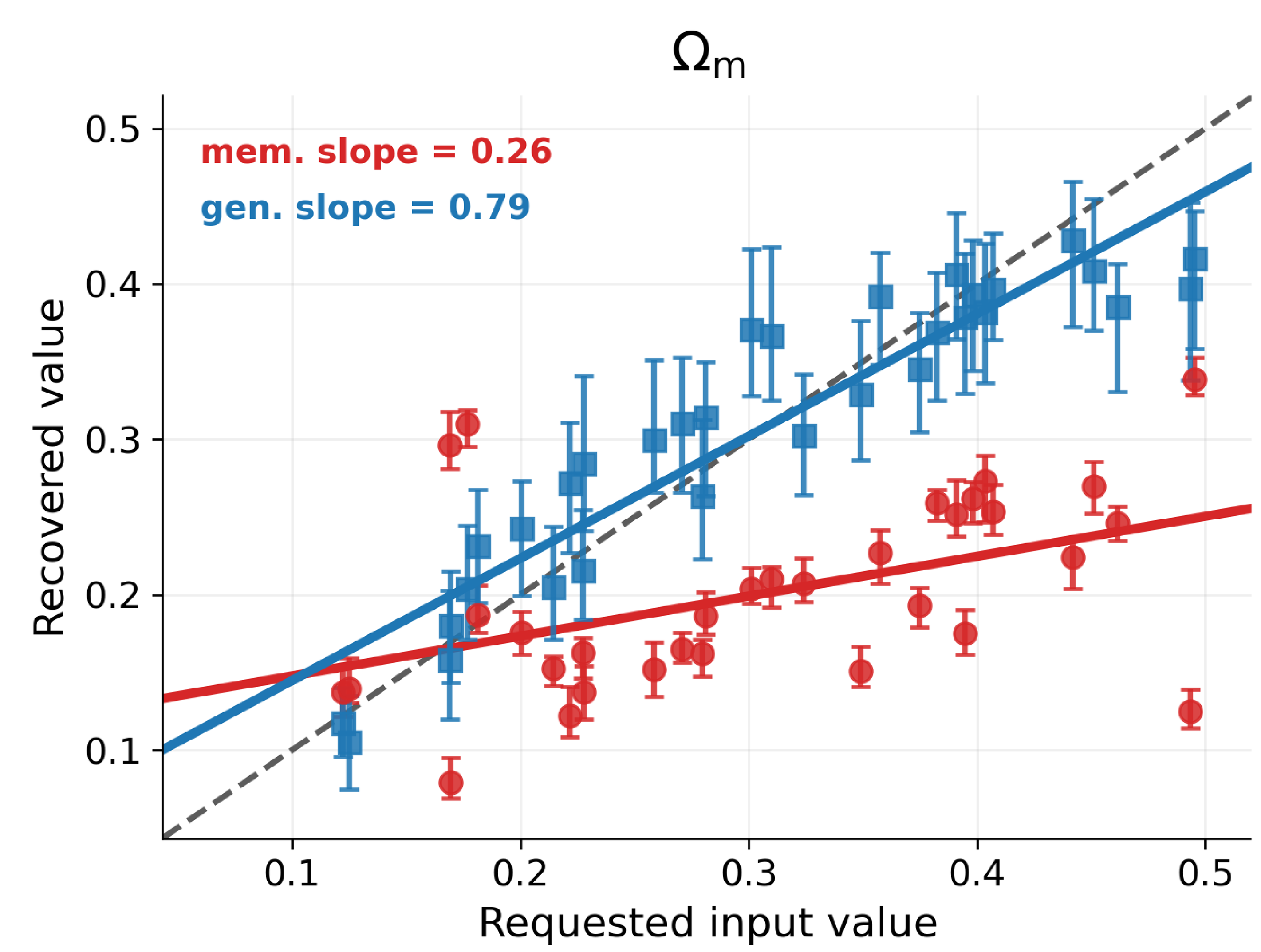
Calibration: does the generator obey θ?

Ask the conditional model for **held-out cosmologies**, then read the cosmology back from its generated maps with a **frozen probe**. A memorized model returns the training distribution no matter what you ask for; a generalizing model tracks the request.



Probe safeguards: VGG16 frozen (ImageNet); regression head trained on *real* slices from non-held-out simulations only; held-out sims 900–931 used exclusively for testing and calibration.

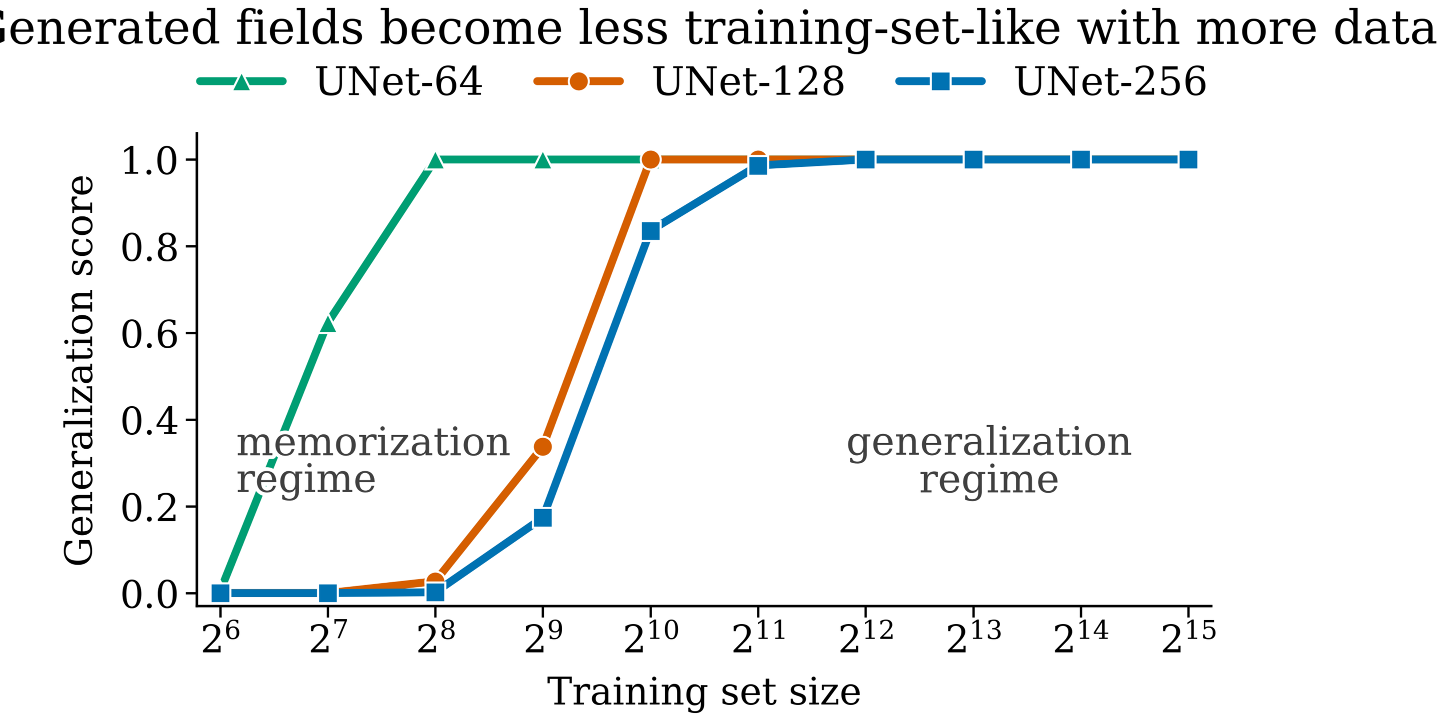
Calibration results



Main result: recovered-vs-requested Ω_m slope $\approx$ **0.79** in the high-data regime vs **0.26** under memorization. The high-data model tracks the requested input with nonzero seed-to-seed scatter, rather than collapsing to one training-like response.

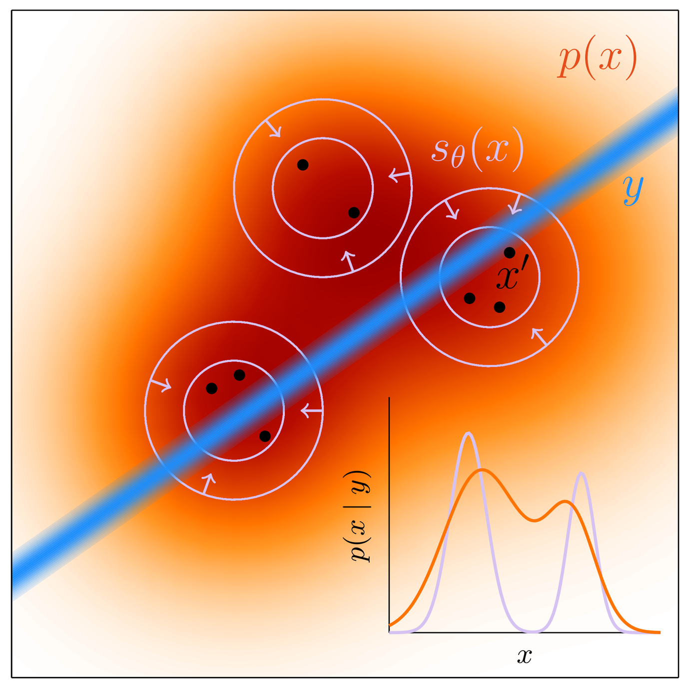
Probe skill on **real** held-out fields (R^2, frozen VGG16 + MLP): Ω_m **0.91**, σ_8 **0.74**, A_{SN1} 0.45, A_{SN2} 0.33, $A_{\mathrm{AGN1}} \approx 0$, A_{AGN2} 0.10 — astrophysical feedback is only weakly imprinted on single HI slices, so calibration leads with Ω_m.

Memorization $\rightarrow$ generalization transition



- Score uses a **real-data calibrated q95 threshold**: higher means generated fields are less training-set-like in PCA space.
- **Small** N_{2D}: generated maps remain close to training fields.
- **Larger** N_{2D}: UNet-64 / 128 / 256 move toward novel samples; the wider UNet-256 transition is shifted to larger data.

Why this matters for inference



Orange denotes the true field distribution, the blue band is the data constraint y, and the black points are training-like islands learned in the memorization regime. A likelihood using this collapsed generator can return a sharp but biased posterior on those islands, rather than the true $p(x \mid y)$. We test this failure mode directly:

$$\boldsymbol{\theta}_{\mathrm{ask}} \rightarrow x_{\mathrm{gen}} \rightarrow \hat{\boldsymbol{\theta}}.$$

Generated maps must preserve the requested cosmology before the generator is trusted as an inference prior.

Takeaways

1. Diffusion models on CAMELS show a clear **memorization $\rightarrow$ generalization transition** with training-set size, located by NN checks, PCA embeddings, SSCD cross-checks, and cross-model reproducibility.
2. Under memorization, recovered cosmology is **tight but biased** toward the training mean; in the high-data regime it **tracks** θ_{ask} with realistic sample-to-sample spread.
3. Summary statistics alone are not enough; a useful emulator must be both **physically faithful** and **calibrated to the requested cosmology** before it is trusted for inference.